

An Affection-Based Dynamic Leader Selection Model for Formation Control in Multirobot Systems

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Abstract—In this paper, a dynamic leader selection process of a multirobot system with leader–follower strategies is studied in terms of formation control. A fuzzy inference system is employed to evaluate the status of robots by means of their states. Based on the status, an affection-based model is proposed to trigger a leader selection module. Followers send out unsatisfied signals when they are disappointed at the current leader. The abashment value of the leader changes with its own status as well as the number of unsatisfied signals received from its followers. When its abashment value goes beyond a given threshold, a leader reselection process is triggered. Moreover, a swap-greedy algorithm is proposed to approximate the optimal solution for confirming the leader–follower relationship, which can be described as a combinatorial optimization problem to minimize the total travel distance of all the robots. Extensive simulation results demonstrate that the proposed model can improve the probability of a robot team escaping from local extreme points significantly, and even in the case of leader failure, the team can reselect a leader autonomously and keep moving toward the target.

Index Terms—Affection model, formation control, leader selection, multirobot system (MRS).

I. INTRODUCTION

IN THE past few years, multirobot systems (MRS), with their distributed parallel task processing capability and simple design structure, have attracted much attention from various research groups. Their cooperative control and consensus problem as a specific instance of a multiagent system (MAS) is

focused due to its wide applications in system biology, control engineering, and social science [1]–[3].

Motivated from the collective motion in biological systems, such as bird flock and mammal migration, a leader–follower strategy is introduced to deal with the cost issue caused by their different computing abilities. Many researchers considered the controllability problem in the structure of MAS with a leader–follower strategy [4]–[7]. Ji *et al.* [5] studied the connection between topology structures of an interconnection graph and controllability of MAS in terms of controllability, and proposed a necessary and sufficient condition based on the eigenvector of subgraphs. Li and Spong [4] proposed a geometric decomposition method to deal with the MAS with both position coupling and velocity coupling, and also answered how to transform a leader–follower system into a leaderless one. Jafari *et al.* [6] discussed a leader–follower MAS with multiple leaders and provided a graphical condition for structural controllability. Olfati-Saber [7] provided a flocking algorithm for a group of agents to track the path of a virtual leader by including a navigation feedback to every follower. Su *et al.* [8] relaxed the assumption to that only a fraction of agents were informed, and extended the algorithm in [7] to adapt to the velocity variation of the virtual leader. Different with using global information, Zhao *et al.* [9] first proposed a smooth control algorithm only based on output measurements to deal with the distributed average computation problem in MAS. Recently, Liu *et al.* [10] and Zhao *et al.* [11], [12] investigated the finite-time consensus problem for second-order nonidentical MRS. In their work, they designed the tracking protocol based on both relative positions and velocity measurements, and also proposed a novel protocol in which only relative position measurements were available [11]. Furthermore, by carefully designing a Lyapunov function initially, the max boundary of settling time was estimated theoretically [12]. Liu *et al.* [10], [13] extended their work to the formation control problem in MAS. They presented an optimal control law to steel the system to the designed formation in a given time, and maintain the minimal control inputs in the meanwhile.

However, there is not a very suitable model to select a robot leader dynamically in the MRS formation control, although several remarkable achievements about leader selection have been made to achieve the MAS state convergence [14]–[17]. In the previous work, some researchers treated positions of agents as states and conversed the formation problems into state

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convergence issues. By ignoring real obstacles, they defined a kind of noise with a certain pattern and selected the leaders' focuses in order to accelerate the convergence rate or decrease the convergence error. But during the converging process, the leader was fixed. Unlike the state convergence in the MAS, for real cases, the probability of encounter obstacles is not equal from the view of the time distribution, thus a system with a previous defined leader may still suffer from leader trapping problem. Thus, the convergence error is not a suitable principle to select leaders for the time-varying situation that the robots face. When the robots move into a dilemma, a fixed leader strategy cannot guide members to go out of woods no matter which leader is selected before. In addition, if the selected robot leader fails during the traveling, the entire team stops. One way to break this standstill is to allow a central controller to reassign a leader [18]. Some empirical studies pointed out that the leadership is greatly related to an individual's need [2], [19]. For example, the nutritional state of individual fish in a school affects its position significantly, and a dynamic game-theoretical model has been employed to simulate the spontaneous emergence of leaders [19]. Meanwhile, other researches attempted to explain a leader's emergence in a view of individual's temperament [2], [20]–[22], but only a few attempts have been made to improve the MRS performance. Therefore, we concentrate on the formation efficiency and autonomous behavior patterns, which make the MRS more efficient, adaptive and easy to implement.

In a human society, an easily observed phenomenon is that affections have important influences on human behaviors, decision-making and interactions among human beings [23]. The cooperation among individuals is effected by the feel itself under an environment. But so far, a specified and generally acceptable ideal emotion definition is still lacking especially when it is applied to the MRS. Researchers have to abstract the affection reaction phenomenon creatively to match their particular problems [24]–[26]. Recently, by introducing the fuzzy inference systems (FISs) into a state-variable emotion model, the subjective and intuitive concepts of human beings were applied to the robot guide [27]. Salichs and Malfaz [28] presented a decision-making system based on drives, motivation, and artificial emotions to satisfy the internal needs of an agent. In their work, each emotion was modeled separately and combined with a learning process to release these artificial emotions adaptively to maximize an agent's well-being. Kusano *et al.* [29] divided the robotic emotions into efficacy and stimulation, and put forward an emotion driven robot dynamic model. In their work, the robots search and avoid obstacles according to several given behavior rules. Lee-Johnson and Carnegie [25] introduced artificial emotions into the mobile robots to improve their adaptive performances for a navigation task.

In this paper, we try to solve the dynamic leader change problem, and propose a self-adaptive DLS model based on two virtual affections. The main differences between our model and previous work are as follows.

- 1) The robots can change their leaders based on environmental situations and their own conditions during their trips.
- 2) When the leaders malfunction, they need not send a specific signal to tell the followers to switch a leader. Instead, the followers can select a new leader autonomously based on their own states without center controllers.

The main contributions of this paper are as follows.

- 1) An affection-based DLS model (ADLSM) is proposed to switch the leaders autonomously and increase the probability for the robots to get out of the woods when a robot team is in dilemma or its leader fails.
- 2) An FIS is employed to build a state evaluator to assess the situations of the robots based on the velocity and environmental conditions.
- 3) A greedy algorithm with a swap operation is proposed to determine the positions of the followers in the formation, which can decrease the travel distance of all robots.

The rest of this paper is constructed as follows. Section II gives a brief introduction to behavior-based formal control models. Section III presents a self-adaptive leader selection model based on an artificial affection mechanism with a highly efficient position-establishment algorithm. Section IV analyses some of the detail results from simulations. The conclusion and future work are given in Section V.

II. FORMATION CONTROL MODEL

In this section, a robot formation structure according to the behavior-based formation generation is employed. A leader–follower strategy is introduced to preserve the specific shape and guide the robots to their destinations.

A. Problem Statement

The main problem we consider is to form a given shape in a limited time, and maintain the formation toward a target location [30]. During this process, all robots should have the following features.

- 1) A robot should avoid collision among all robots while traveling.
- 2) A particular robot, the leader, knows the exact position of a given target and moves toward it. The other robots, called the followers, follow the leader and keep a specific distance and angle with the leader, such that the formation maintains.
- 3) A robot should split to avoid obstacles and rejoin the formation after crossing a complex terrain.

In addition, there are a few more general assumptions to be made for the model.

First, each robot, notated as a little circle, could be described by three main characters.

- 1) Its head direction, $\theta_i \in [-\pi, \pi]$ relies on an arbitrary but fixed global reference system, representing its facing direction.
- 2) Its velocity v_i is defined only at the heading direction.
- 3) The minimal distance between two robots is $d_{\min}$, which represents the size of robots. The positive value means that the robot is moving forwards, whereas the negative one means backwards.

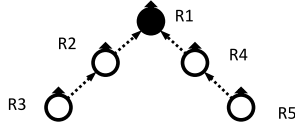


Fig. 1. Example of V-type formation. Leader is emphasized by dark color.

Second, the robots can detect other objects in a limited region near them.

Third, since our main task is to identify the dilemma and change the leader autonomously, in order to simplify the task, we assume that the communication among robots is unlimited, which means there exists a link between any two robots.

Lastly, a formation matrix gives the formation in advance, and the leader robot may change it.

B. Formation Description

To show a robot team under a leader–follower strategy [31], we give an example of V-type formation with five robots, i.e., R_1 – R_5 , as shown in Fig. 1. R_1 is defined as the global leader, i.e., it knows the exact position of the target, and other robots as the followers. R_2 and R_4 follow R_1 directly, R_3 and R_5 follow their teammates, R_2 and R_4 , respectively, and accordingly follow R_1 indirectly. During this formation principle, a robot is a unique leader to other robots, and follows its own leader except R_1 . In more details, at each time step, a robot could only have one leader, but it could guide as many as the design needs. So the formation control at a team level can be simplified to a series set of formations with two robots. Since only the followers take the responsibility to maintain the formation, and the team formation can be guaranteed completely if every follower in each leader–follower pair is allocated at its accurate position with respect to its leader, the communication requirement among robots can thus be greatly reduced.

Consider the case of only two robots, i and j , robot i as a follower follows robot j as a leader. While traveling, they should have a consistent heading direction, and robot i keeps a constant distance with robot j . In order to avoid complex communication, the information, which each robot can easily obtain, is utilized to build the formation, such as the position and velocity of the leader.

The main symbols used in this paper are given in Table I.

Here, we limit all angle variables at the range of $[-\pi, \pi]$, and $\Delta\psi_i = \theta_i - \psi_{ij}$. We use a formation matrix to describe the designed formation with N robots as follows [30]:

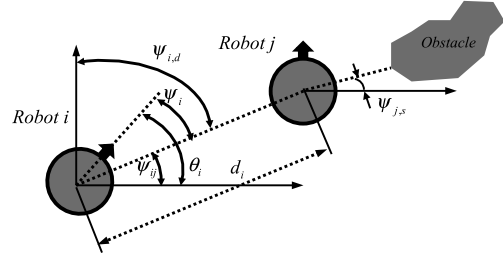
$$S = (L, \psi_d, l_d) \quad (1)$$

where $L_{i,d} \in L$ is the leader's ID of robot i . $L_{i,d} = 0$ means robot i is the global leader. $\psi_{i,d} \in \psi_d$ is the heading error between robot i and the designed direction, and $l_{i,d} \in l_d$ is the designed distance between leader i and its follower. Take Fig. 1 as an example. Its formation matrix is

$$S = \begin{pmatrix} 0 & 0 & 0 \\ 1 & \pi/4 & 0 \\ 2 & \pi/4 & 0 \\ 1 & -\pi/4 & 0 \\ 4 & -\pi/4 & 0 \end{pmatrix}. \quad (2)$$

TABLE I
MAIN SYMBOLS

Symbol	Representation	Unit
$Robot\ i$	The follower robot	-
$Robot\ j$	The leader robot	-
θ_i	The heading direction of robot i	rad
θ_j	The heading direction of robot j	rad
$\theta_{i,d}$	The ideal heading direction of robot i decided by robot j	rad
ψ_{ij}	The direction of robot j at robot i 's position	rad
$\psi_{i,s}$	The direction of obstacle s at robot i 's position	rad
$\psi_{i,d}$	The designed error between robot i 's heading direction and robot j 's position	rad
ψ_i	The actual error between robot i 's heading direction and robot j 's position	rad
$d_{i,j}$	The distance between robots i and j	m

Fig. 2. Illustration of notation with two robots and an obstacle. Robot i is supposed to be the follower and robot j is the leader.

Among S , $L_1 = 0$ means R_1 is the global leader. $L_2 = 1$ means R_2 follows R_1 . It has to be noticed that at the beginning, we fix L in advance, but it may vary depending on robots' states when the team falls into trouble. Our purpose is to converge ψ_i to $\psi_{i,d}$ and v_i to v_j with a constant $d_{i,j}$. Fig. 2 shows an example for two robots. Kinematic models for heading direction and velocity of robot i taking robot j as a reference point are defined as follows:

$$\dot{\theta} = f_{\text{approach}}(\theta_i) + f_{\text{divert}}(\theta_i) \quad (3)$$

$$\dot{v}_i = -\alpha_i(v_i - v_{i,d}). \quad (4)$$

Equation (3) consists of approaching and diverting terms

$$f_{\text{approach}}(\theta_i) = \frac{\lambda_{\text{app}}}{1 + e^{-(l_{i,d} - d_{i,j})}} \sin(\theta_i - \psi_{ij} - \psi_{i,d}) \quad (5)$$

$$f_{\text{divert}}(\theta_i) = \frac{\lambda_{\text{div}} e^{-(l_{i,d} - d_{i,j})}}{1 + e^{-(l_{i,d} - d_{i,j})}} \sin(\theta_i - \psi_{ij} - \psi_{i,d}) \quad (6)$$

where λ_{app} and λ_{div} is the scale factor to regulate the approach force and divert force. The desired velocity, $v_{i,d}$ in (4) is calculated by

$$v_{i,d} = \begin{cases} v_j - (l_{i,d} - d_{i,j})/T & \text{if } d_{i,j} \geq l_{i,d} \\ -v_j - (l_{i,d} - d_{i,j})/T & \text{else} \end{cases} \quad (7)$$

where parameter T is used to smooth the movement of robot i .

Equations (3)–(7) can guarantee the robots to move in a designed formation, but the robots need to avoid obstacle collisions during the trip. In the obstacle avoidance behavior,

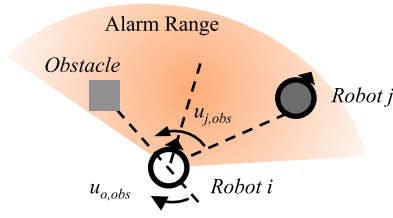


Fig. 3. Illustration of a robot when there exist obstacles in its alarm range. Robot j can be either a follower or a leader.

a robot treats the others as dynamic obstacles as well. By applying the repulsive force method, we only use the easily obtained measurements to calculate the angles between robots and obstacles. These data can be got by the embedded sensors of a robot, infrared and ultrasonic sensor for instance. Denote $\psi_{i,obs}$ as the angle between robot i and an obstacle, the repulsive force $U_{i,obs}$ is defined as

$$U_{i,obs} = \lambda_{obs} \sum_s (\theta_i - \psi_{i,s}) e^{-\frac{(\theta_i - \psi_{i,s})^2}{2\sigma^2}} \quad (8)$$

where λ_{obs} is the scale factor of the repulsive force and σ is used to control the turning sensitivity of robots. A bigger σ means a larger detection range. Thus, we rewrite (3) as follows:

$$\dot{\theta}_i = \gamma \sum_k f_k(\theta_i) + U_{i,obs}(\theta_i) + \xi \quad (9)$$

where $k \in \{\text{approach}, \text{divert}\}$, ξ is a designed stochastic disturbance, which is used to avoid the zero point of (8), and γ is an index factor. It takes value 0 when there is an obstacle in the collision alarm range of robot i , otherwise it takes 1. The alarm range of collisions, as shown in Fig. 3, is designed in advance to prevent damage of the robots. It is smaller than the detection range of the robots.

In the case of Fig. 3, $u_{j,obs}$ is the moment of force caused by robot j , and $u_{o,obs}$ is the one caused by the obstacle. We can see that the angle between two robots is smaller than that between robot i and obstacle, thus $u_{j,obs}$ is larger than $u_{o,obs}$ depending on (8). Robot i spins anticlockwise in order to avoid a collision with robot j .

C. Stability Analysis

The MRS consists of two parts: 1) the velocity part, which is a linear system as shown in (4) and 2) the heading direction part, which in the nonlinear system as shown in (9). Based on linear stability theory, the velocity subsystem is stable for an arbitrary $\alpha_i > 0$, since $(d\dot{v}_i/dv_i) = -\alpha_i < 0$. For the nonlinear subsystem, we consider the following Lyapunov function for each robot:

$$V(\theta_i) = \gamma V_{\text{approach}}(\theta_i) + \gamma V_{\text{divert}}(\theta_i) + V_{i,obs}(\theta_i) + K \quad (10)$$

where $K \in \mathbb{R}$ and

$$V_{\text{approach}}(\theta_i) = -\lambda_{\text{app}} E \cos(\theta_i - \psi_{ij} - \psi_{i,d}) \quad (11)$$

$$V_{\text{divert}}(\theta_i) = -\lambda_{\text{div}} (1 - E) \cos(\theta_i - \psi_{ij} - \psi_{i,d}) \quad (12)$$

$$V_{i,obs}(\theta_i) = \lambda_{obs} \sum_s \sigma^2 e^{-\frac{(\theta_i - \psi_{i,s})^2}{2\sigma^2}} \quad (13)$$

where

$$E = 1 / (1 + e^{d_{i,j} - l_{i,d}}). \quad (14)$$

It is easy to tell that $V(\theta_i) - K$ is bounded. Thus, we have $V(\theta_i) > 0$, for any K satisfies the following condition:

$$K > \lambda_{\text{app}} + \lambda_{\text{div}}. \quad (15)$$

Next, since

$$\dot{V}(\theta_i) = \frac{dV(\theta_i)}{dt} = -[\dot{\theta}_i]^2 < 0 \text{ if } \theta_i \neq \tilde{\theta}_i \quad (16)$$

where $\tilde{\theta}_i$ is the zero point of (9). Thus the heading direction system is asymptotically stable in the sense of Lyapunov.

III. AFFECTION-BASED LEADER SELECTION MODEL

Since the leader is always fixed with a leader-follower strategy, the initial states of robots may affect the performance of formation. Besides, while they are travelling, the leader may cause the whole team unable to move if it is trapped or fails. Thus, in this section, inspired by a concept of human affection requirement, we introduce an ADLSM to the MRS. The proposed model aims at improving the efficiency of formation and the robustness of team moving.

A. Concept and Model of Affection

There exists such a phenomenon in a society, a leader is not always fixed in a group. When an individual is in a comfortable state in the environment, she/he tends to give her/his leader a relatively high evaluation, and thus inclines to follow such leader. When an individual's living environment deteriorates, she/he becomes more anxious and disappointed with the current leader. The leader feels guilty, even resigns when she/he leads a team into a difficulty or dilemma. A new and perhaps better one may emerge or be selected to replace the original leader and guide the entire group.

An MRS faces the similar problem when it runs in a complex environment. Once swarm robots move into a dilemma with a formation constraint, the leader surrounded by obstacles may be trapped and unable to move. If other robots still follow the trapped leader, the entire team will be in trouble and cannot move toward their target. Therefore, a new and hopefully better leader should be selected such that it can lead the team out of trouble, thereby reasonably and dynamically making the MRS to adapt to unknown environment. In order to learn from the leader selection phenomenon in biological population, the following definitions to measure the psychological state of robots need to be specified.

Definition 1 (Disappointment): The disappointment of a robot is an abstract evaluation of its state. It increases with the duration of a robot in a dilemma. The higher disappointment, the intenser an individual is to abandon the current leader and selects a new leader. When the disappointment value of an individual exceeds a certain threshold, the follower will broadcast a signal to the group leader to ask for a reselection. The disappointment value is calculated by the following formula with a range of $[0, 1]$:

$$D_i(t) = \min(D_i(t) \times e^{-1/T_i}) + \delta(1 - e^{-k_d(t-\tau)}), 1 \quad (17)$$

where T_i is the half-life variable, k_d is the integration constant and τ is the moment when robot i traps into a dilemma. δ takes 1 when the robot is trapped, otherwise takes 0. Equation (17) denotes that the disappointment value of robot i , D_i , eliminates as time going on [the first item of (17)], and enhances when the robot is trapped. The second item of (17) will reach 1 if a robot is trapped for a while. In such situation, the recursive calculation of D_i may beyond 1. To keep D_i in the range of $[0, 1]$, we take a min-operation.

Definition 2 (Abashment): The abashment is the unique attribution of the leader, and it increases with the lasting time that the leader has stranded in the plight and the number of reselection signals it has received. The abashment with the range of $[0, 1]$ is denoted as follows:

$$A_i(t) = \min\left(A_i(t) \times e^{-1/T_i} + \frac{1}{k} \left[\delta(1 - e^{-k_d(t-\tau)}) + \frac{N_i}{N} \right], 1\right) \quad (18)$$

where N represents the total number of swarm robots in the communication range. When the leader cannot receive any signal from some of its followers, N decreases. N_i is the number of the robots who ask for reselection a new leader at present, and k is the scaling factor to regulate the value of A_i . The rest parameters have the same meaning with those in (17). When the abashment exceeds a designed threshold, the leader abandons the leadership and initiates a reselection process.

T_i , k_d , and k_a in (17) and (18) determine the rising slope of emotion values. It goes sharply or gently when $1/T_i$, k_d , and k_a increase or decrease. Due to the influence of these parameters, the MRS switches its leaders after a period of time when the robots drop into trouble, and this time delay is indispensable for avoiding too fast leader updating.

It has to clarify that the two affections defined above are only used for stimulating the leader selection process. They do not indicate any physical states of the robots.

Now, we need to define the current status (CS) of a robot to trigger the affection model. The following factors are taken into consideration when evaluating its status.

- 1) Whether the robots stay close to the target (nearby).
- 2) The density of obstacles surrounding a robot (density).
- 3) The velocity of robot i , v_i .

Wherein factors (1) and (2) are used to describe the state of the environment obtained from the sensing system attached on the robots. By reading the axle laser encoder, factor (3) can be easily obtained too. In order to ensure the calculation results in the same order of magnitude, we normalize factors (2) and (3) as follows:

$$I_{\text{density},i} = \begin{cases} N_{\text{obs}}/N_{\text{obs}}^*, & \text{if } N_{\text{obs}} < N_{\text{obs}}^* \\ 1, & \text{otherwise} \end{cases} \quad (19)$$

$$I_{v,i} = v_i/v_{\text{max}} \quad (20)$$

where N_{obs}^* is the predefined maximal number of obstacles in its sensing range, and v_{max} is its permitted maximum velocity.

These factors can be expressed parametrically as a Boolean value and two continuous variables in $[0, 1] \times [-1, 1]$. We employ an FIS to draw the status with different values of factors with fuzzy level of well, risky, and dilemma. Factors (2) and (3) take values from fuzzy sets {empty, sparse,

TABLE II
FUZZY RULES IN THE FIS

No.	I_{density}	I_v	I_{Nearby}	CS
1	-	-	true	well
2	sparse	NS	true	risky
3	intensive	NS	true	dilemma
4	-	FA	true	well
5	empty	NF	true	dilemma
6	empty	FA	false	well
7	~intensive	NF	false	well
8	sparse	~ST	false	risky
9	-	ST	false	dilemma
10	intensive	-	false	dilemma

* “~” used in the table means “not” and “-” means “not given”.

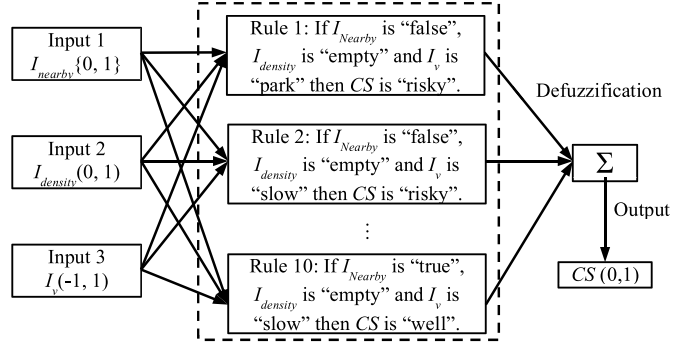


Fig. 4. Basic structure of the FIS for robot's CS.

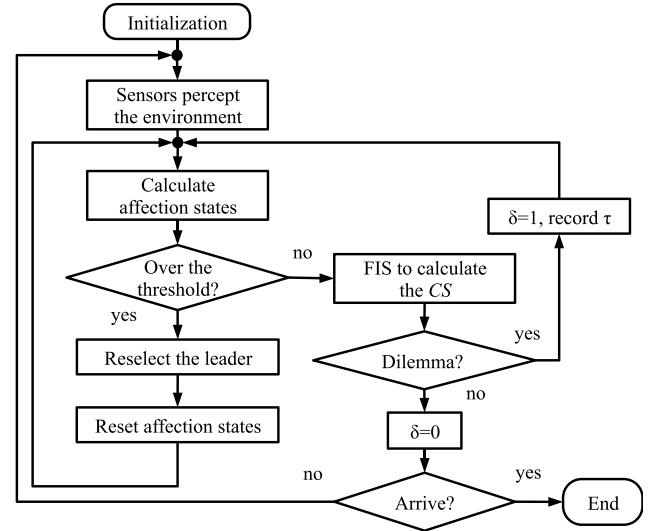


Fig. 5. Flowchart of the affection model.

intensive} and {NF, NS, ST, PS, PF} on the basis of membership functions where NF and NS stand for “negative fast” and “negative slow,” ST means “static,” PS and PF are “positive slow” and “positive fast,” respectively. We use the center area defuzzification method to obtain the view of status surface. The FIS structure with ten rules is shown in Fig. 4. The fuzzy rules are listed in Table II. Thus the leader reselection process of the MRS based on the affection trigger can be represented by the flowchart in Fig. 5.

When the robots travel, they calculate their own affections as one of their properties. When they are in a complex terrain,

the CS will overpass the threshold defined in advance. This triggers the integral process of the two affections. When the global abashment value of the leader exceeds the threshold, the reselection occurs. The new leader and movement strategy will be utilized to help the team out of trouble.

The main idea of the reselection process is to select a new leader who has the lowest CS value. Under the assumption of unlimited communication, a robot can receive all CS values of the others, and the one who has the smallest CS value will win the election.

B. Swap-Greedy Algorithm to Minimize Trip Distance

According to the above discussions, a state dominated robot could be selected as a new leader, but another serious problem is how to confirm the leader–follower relationships of all the other robots. In another word, we need to determine the L in the formation matrix S [see (1)]. An acceptable objective is to minimize the whole trip distance while robots build the formation. Hence, we can formulate the following combinatorial optimization problem:

$$\begin{aligned} \underset{x}{\text{minimize}} \quad & J(x) = \sum_{i=1}^{N-1} \|X_i - P(x_i)\| \\ \text{subject to} \quad & x_i \neq x_j, \text{ if } i \neq j \end{aligned} \quad (21)$$

where, J is the total trip distance of all robots, $X_i \in R^2$ is the coordinate of robot i , $P(x_i)$ is the set of virtual locations referring to the new leader, and the index $x_i \in \{1 : N - 1\}$ is the optimization variable.

However, even in the circumstance that there exists a central controller knowing all the global information, the computation complexity of solving problem (21) is $O(n!)$. Although an offline intelligent algorithm can afford a better solution via strategically searching the solution space, the time cost for iteration makes it unsuitable for the dynamic problem that has a strict real-time requirement. Thus, a fast heuristic formation generation algorithm to solve the problem is required.

Considering the real-time changing of a robot's position and the efficiency requirements of the algorithm, the optimal solution in a dynamic environment changes frequently and requires many computing sources to obtain. Therefore, we propose an improved swap-greedy algorithm to determine the upper bound of J , as shown in Table III. This algorithm successively determines the corresponding virtual locations of the followers in a numerical order. A robot first selects the nearest unlabeled virtual location, then a comparison and swap operation are executed after obtaining J . The robot exchanges its position with a robot determined before, then compares the value of J . If J is decreased, two robots exchange their positions. The robot that is swapped out repeats this comparison until the last replaced one determines its own virtual location. After that, a new undetermined robot joins into this procedure. As can be seen, this algorithm always converges toward the direction of decreasing J . For the optimal index x^* corresponding to J^* , it must satisfy the condition $J^* = J$. That is, the optimal solution for problem (21) is included in the algorithm strategy. Therefore, this algorithm is feasible.

TABLE III
PSEUDOCODE OF THE SWAP-GREEDY ALGORITHM

ALGORITHM 1. SWAP-GREEDY FOR FORMATION GENERATION	
1:	When initialization, the new leader broadcasts the virtual location determined by its location and heading direction. Set all virtual location flags (F_{VL}) to 0. All the followers are classified into a set called “undetermined”, denoted as U . Initialize an empty set called “determined”, denoted as D .
2:	For each robot R_i in U , find the nearest virtual location (VP_i) whose F_{VL} is 0;
3:	Set $F_{VL} = 1$;
4:	Calculate $J_{temp} = \sum_{j=1}^i \ X_j - P(x_j)\ $;
5:	$S = \ D\ \% \ D\ $ is the number of elements in D ;
6:	If $S > 0$
7:	Perform a swap operation;
8:	Endif
9:	Put R_i into D ;
10:	$J = J_{temp}$;
11:	Endfor

TABLE IV
SWAP OPERATION

ALGORITHM 2. SWAP OPERATION	
1:	For each s in S
2:	Swap the selected virtual locations of robot i and s ;
3:	Calculate J ;
4:	If $J \leq J_{temp}$
5:	Swap confirmed;
6:	$J_{temp} = J$;
7:	Else
8:	$s++$
9:	Endif
10:	Endfor

Furthermore, the worst case is that each newly added robot leads to a swap operation for all the robot determined before, thus resulting in the complexity $O(n^3)$. It is much lower than $O(n!)$ in the case of a large number of robots. For the best case, each newly added robot exchanges its position with the one of the first robot. That leads the complexity of the algorithm to $O(n^2)$. The pseudocode of Algorithm 1 is as shown in Table III. The swap operation is described as in Table IV.

C. Parameter Analysis

For the affection model, there are many parameters that have impact on the frequency of leadership reselection, and can affect the MRS performance. When the reselection frequency is too high, the operating cost increases sharply. Meanwhile, after switching a leader, the followers have to change their travel directions in order to track the position of the new leader. All the disadvantages usually deviate the entire team from a target point. Too much switching in the MRS will also make the team stagnate or wander back and forth around a certain position. The influence of several key parameters on the MRS performance is analyzed to ensure that the MRS switches leaders in a reasonable frequency.

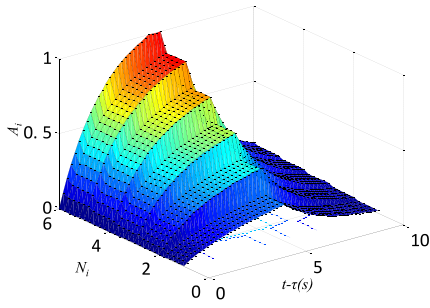


Fig. 6. Surface view of abashment. Take k_a as 0.02 and τ as 0. The CS changes to “well” at 5 s.

The parameters that influence a leader switching process include the status threshold to stimulate (17), half-life and integration constants in the affection model, scaling factor in (18), and the critical point to trigger reselection. Determining a set of reasonable parameter values is a combinatorial optimization problem. However, three main types of parameters, i.e., those from environmental perception, affection calculation, and leader reselection, are independent. Most of them can be given according to the experimental results and human experience except the status threshold and the scaling factor. The former determines the start time of integration in the affection model, while the latter affects the accuracy of the calculated abashment value.

One of the most vital parameters is the status threshold. It has an impact on the moment the affection model starts to work. Therefore, the status of a robot in the vicinity of the threshold must be transient, and uninterruptible jitter (i.e., changing state) is not allowed in this area. A set of experiments are conducted to determine the robust range of the status threshold, in which the CS of robots makes the minimal number of shuttling times.

Another one is the scaling factor k in (18). If k takes a too small value, the abashment value will always equal 1, but an overlarge k leads to the low value of abashment, which cannot trigger any reselection. As can be seen from Fig. 6, the abashment is a discrete function, but under the circumstance that N_i takes a certain value, it is a piecewise continuous function. Since the trigger of leader reselection must have happened in the rising edge of the abashment curve (abashment reset to 0 after reselecting a new leader), and the trigger threshold is less than 1, the update model (18) can be converted into the following form:

$$A_i(t) = \frac{1}{k(1-B)} \left[1 - e^{-k_a(t-\tau)} + N' \right] \quad (22)$$

where $B = e^{-1/T_i}$ and $N' = N_i/N$. The value of k is computed as

$$k = \frac{1}{A_{i,\text{trg}}(1-B)} \left[1 - e^{-k_a(t-\tau)} + N' \right]. \quad (23)$$

IV. SIMULATION RESULTS

In this section, a delivery mission via MRS is indicated to test the performance of our model. In this mission, all robots

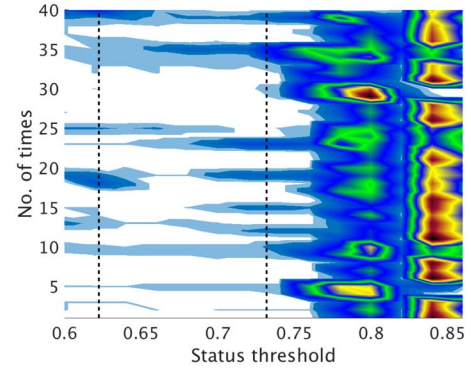


Fig. 7. Robust area of status threshold.

TABLE V
STATISTICS DATA FOR STATUS THRESHOLD

	Low boundary		Up boundary	
	Mean	Std	Mean	Std
Status threshold	0.6255	0.0357	0.7315	0.0477

pick up goods in a certain area. Since the storage points are randomly distributed in this area, robots will be initiated randomly. After carrying their goods, they have to form a specific formation, “V-type” for instance, due to the energy saving or safety consideration. Then they need ship goods to a given destination while maintain the formation. Robots need to deal with two kinds of challenges: 1) semicircle-shaped obstacles may trap them and 2) the leader may be failed or destroyed during their trip.

In order to apply our model to finish this mission, we first employ numerical simulations to determine the parameters of the ADLSM. After that, a series of experiments are employed to test the optimization ability of the proposed swap-greedy algorithm. Then the MRS with the ADLSM and other leader-follower strategies are compared in terms of formation error, formation time-cost (TC), and target arriving rate.

A. Parameter Determination of the ADLSM

As discussed before, the status threshold has a great effect on the leader reselection. To determine its robust range, we introduce the definition of robust range of the CS. It can be decided by the upper and lower bounds. Each value in the set corresponds to the least number of the CS fluttering times.

We apply a mobile robot running in a fixed complex environment for 40 times with the FIS mentioned before. 40 groups of time series data about the robot’s CS are collected and analyzed. Since the FIS starts to output a “dilemma” only after the CS is higher than 0.6, we choose 0.6 as the baseline of status threshold and increase it by the step of 0.02–0.84, which absolutely gets a dilemma from the FIS’s output. The number of shuttling times of the CS under different thresholds is counted to gain the robust range of the CS. In Fig. 7, the number of times across different thresholds is color-coded. The lighter the better robustness threshold value is. The area between two black dotted lines corresponds to a robust range. The upper and lower bounds of status threshold are shown in Table V.

TABLE VI
VALUES OF KEY PARAMETERS

Parameter	$T_{l,d}$	$T_{i,d}$	k_d	k_d	N_{obs}^*	v_{max}	$A_{l,rg}$
Value	100	2	0.01	0.0008	7	5	0.6

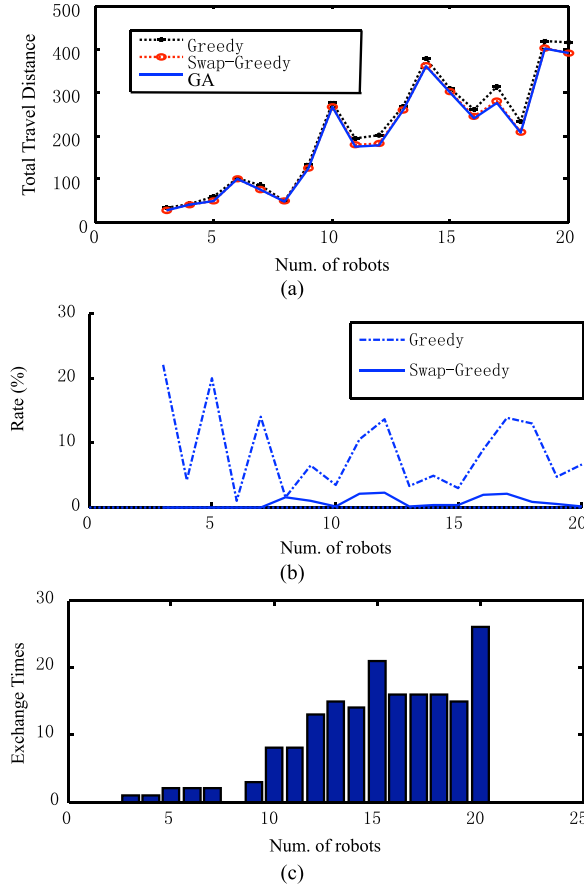


Fig. 8. Trip distance comparison. (a) Calculated by the greedy algorithm without swap operation, swap-greedy algorithm, and GA, respectively. (b) Relative deviations of the greedy algorithm and the swap-greedy algorithm. Rate = $(J - J_{GA})/J_{GA} \times 100\%$. (c) Exchange time of the swap-greedy algorithm.

To minimize the oscillation, we select the average value of the robust area as the status threshold, thus leading to threshold = 0.69325.

Consider the situation that a reselection happens when over 3/4 members in the team send a despair signal to the leader with its abashment over 0.6. According to (23), the scaling factor is $k = 3.189$. Other parameter values are listed in Table VI.

B. Approximate Solution of Optimal Virtual Location

As a heuristic algorithm, the proposed greedy algorithm can only guarantee a local optimal solution. A comparison experiment has been performed to examine the convergence ability of the proposed algorithm to approximate the optimal solution of the followers' virtual locations. In the experiment, all robots are randomly initiated in the environment. The number of robots involved in the experiment varies from 3 to 20. Every time one specific robot is randomly selected as the leader. The followers generate an annular formation based on the

TABLE VII
RELATIVE ERRORS

Alg.	Greedy/GA (%)	Swap-Greedy/GA (%)
Ave	8.55	0.70
Std	6.25	0.84

leader's position and direction. For comparison, a genetic algorithm (GA) with population size being 100 and 200 iterations is used to solve the same problem. The comparative results are as shown in Fig. 8.

It is observed from Fig. 8(a) that the GA finds better solutions (i.e., closer to optimal ones), but a global searching algorithm needs too much TC and information of all robots. The relative deviations given in Fig. 8(b) demonstrate that the greedy algorithm is also feasible to solve problem (14). Table VII indicates the relative errors of the greedy algorithm and swap-greedy algorithm with respect to the GA. The swap-greedy algorithm can limit the relative error less than 1% but much faster than the GA.

In conjunction with Fig. 8(c), our results show that the swap-greedy algorithm has a better optimization capability to confirm the relationship with an acceptable complexity growth. Thus the solution obtained by the swap-greedy algorithm is employed as the approximately optimal solution.

C. Robustness to Environmental Disturbances

In the following two sections, we will apply our model to the delivery mission and present our main results through computer simulation. Trajectories of robots are given in the form of curves in the x - y plane in each experiment.

Before we start simulation, several characters need to be declared to measure the performances of the formations. First, the alignment [32], [33] is defined by

$$\alpha = \frac{1}{N} \sqrt{\left(\sum_{i=1}^N \cos \theta_i \right)^2 + \left(\sum_{i=1}^N \sin \theta_i \right)^2}. \quad (24)$$

It is used to measure the extent to which the robots are moving in the same direction. When all robots face to the same direction, α becomes 1. It decreases with the directions from the ordered to chaotic. The extent of formation error [30] is defined as the mean square error of all robots to the corresponding virtual locations

$$d = \frac{1}{N} \sum_{i=1}^N \sqrt{(x_l - x_i - l_{i,d} \cos \eta)^2 + (y_l - y_i - l_{i,d} \sin \eta)^2} \quad (25)$$

$$\eta = \theta_l - \psi_{i,d}$$

where x_l , y_l , and θ_l represent the global leader's position in X - Y plot and the heading direction, respectively. x_i , y_i , and $\psi_{i,d}$ are the position and designed difference between heading direction and global leader's position of robot i , respectively. Other variables such as $l_{i,d}$ and $\psi_{i,d}$ can be seen as parameters in a new formation matrix where each robot takes the global leader as a reference point as shown in Fig. 9.

To evaluate the MRS's robustness to external environmental disturbances, the following experiments are carried out.

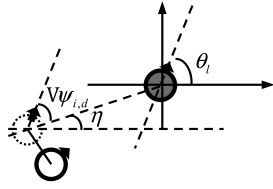


Fig. 9. Illustration of formation error calculation. The darker circle denotes the global leader, and the circle with dotted line is the virtual location where robot i should be. Formation error for robot i is the Euclidean distance between the actual position and virtual one.

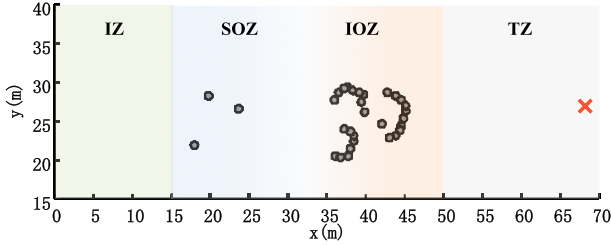


Fig. 10. Complex map for a comparison experiment. The solid spots are obstacles and the red "X" is the target position.

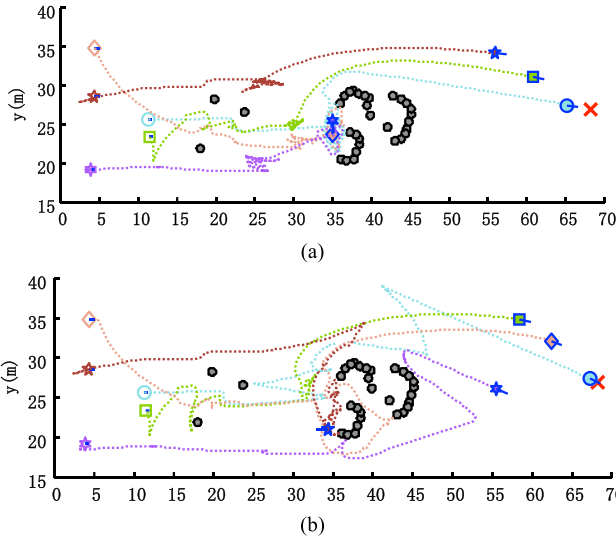


Fig. 11. Trajectory of an MRS. Robots move (a) without changing a leader and (b) with the ADLSM.

According to the environmental obstacle distribution, we build a complex map that consists of initial zone (IZ), sparse obstacles zone (SOZ), intensive obstacles zone (IOZ), and target zone, as shown in Fig. 10. All the robots locate at IZ at the beginning. For accomplishing the delivery mission, they have to deal with SOZ first, which is designed to test the normal obstacle avoiding ability. Then the robot team faces IOZ, formed as several semicircles. Robots are immersed in the local extreme points easily since the potential fields in various directions are superimposed. Fig. 11 demonstrates the moving traces with and without the ADLSM in the complex map. The position errors and consistency of angles are given in Figs. 12 and 13, respectively.

As shown in Fig. 11, the performance of two strategies has no difference while facing SOZ, but varies when dealing with IOZ. Two curves coinciding with each other from the

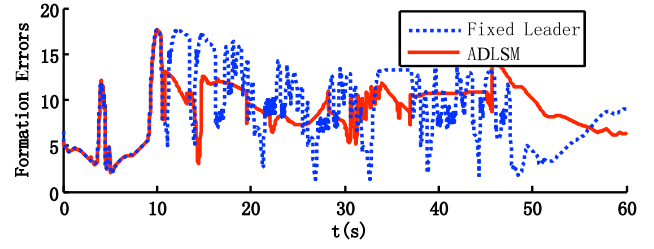


Fig. 12. Formation error comparison.

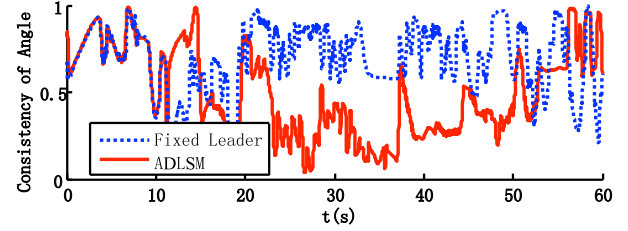


Fig. 13. Angle comparison.

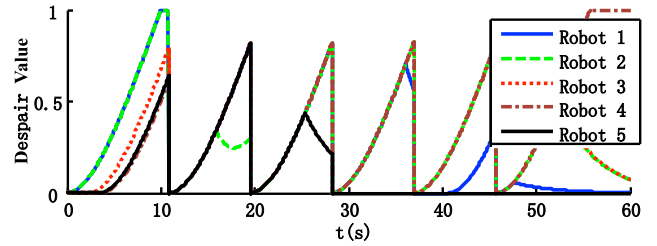


Fig. 14. Despair evolution curves for the ADLSM.

beginning to about 11 s in Figs. 12 and 13 show the secondary evidence of this phenomenon. This is because in the case of SOZ, the environment that robots perceive is not so awkward, and thus no reselection operation is triggered. When they move into IOZ, the formation error of a fixed leader model presents a periodical variation. It also shows the same frequency oscillation in consistency of angles at the same time, which means that most robots are parked but a few standing around. This is mainly attracted by some local minima. The MRS with the ADLSM prefers a lower consistency of angle in IOZ.

Figs. 14 and 15 show the despair and abashment evolution processes of the MRS. At the first few seconds, all the five robots satisfy their status but they are followed by increasing despairs that reach the top from 5th to 11th second. The leader's abashment shows a delayed jump at near the 8th second. This is mainly due to the ascending number of the despaired robots. After reaching the top of the leader abashment at the 11th second, the reselection operation is triggered, and the ADLSM is reset. Switching a leader causes more direction selection to gain a greater probability of escaping from IOZ. During the whole experiment process, the leader reselection occurs for five times and the corresponding leader switching process is as shown in Fig. 16.

To verify the above conclusion, we carry out 40 times of repeated experiments and employ another three different models for comparison. First, we follow assumptions of [14] and [15] treating the MRS as the MAS with zero-man white noise process. Under this condition, a leader is

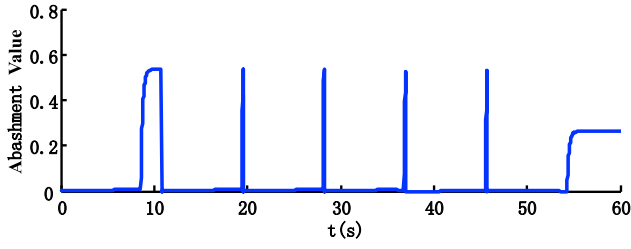


Fig. 15. Abashment evolution curves for the ADLSM.

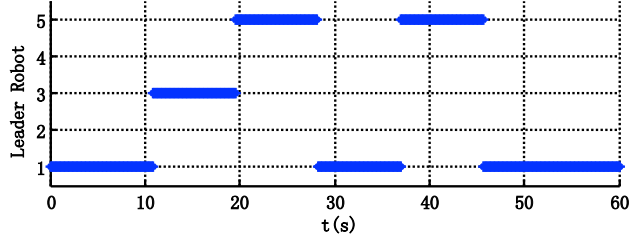


Fig. 16. Illustration of leader switching with the ADLSM.

determined in advance based on their work to minimize the mean-square error (MSE) of convergence. For the sake of clearness, we name this scenario as min-MSE with fixed leader (MFL). Second, we select a robot as a leader randomly after a fixed period of time. We name this method as randomly leader selection (RLS). In the third method, we use our affection model to trigger the leader selection process, but select a leader based on min-MSE principle. We called this method as affection model-MSE leader selection (AM-LS). But in this method, the system needs a central controller to select a leader.

In order to apply the MFL and AM-LS, we introduce the convergence error defined in [14]

$$R(S) \triangleq \lim_{t \rightarrow \infty} \mathbb{E} \|x_f(t) - x_f^*\|_2^2 \quad (26)$$

where $x_f(t)$ denotes positions of followers, x_f^* is the final positions. A formation matrix is determined to minimize R .

In each experiment, all robots are initialized in IZ randomly, and the simulation time is limited to 100 s. We count the heads of robots that reach the target successfully during each experiment. The arrival rate (AR) is calculated as the number of arrived robots divided by the total number of robots. Fig. 17 shows comparisons of AR and TC distributions among these four models. The switch time (ST) period of RLS is 20 s.

The bar chart in Fig. 17(a) demonstrates that the robot team with the ADLSM has a significant decrease in probability of being trapped (AR = 0) in the environment. In contrast with other strategies, the AR distribution of the MRS with the ADLSM concentrates in the high AR area. The comparison of TC to arrive at the target is presented in Fig. 17(b). Different with AR = 0 in Fig. 17(a), TC equals to 100 means that no robot reaches or is close to the target position in the given time. From the bar chart we can see only a few times of experiments with the ADLSM costing 100 s. This result also shows a strong ability for robots to get out of the woods with the proposed model. Another intuitive conclusion is the distributions of the MRS with the ADLSM are more concentrated. To verify this conclusion, we give a statistic comparison among

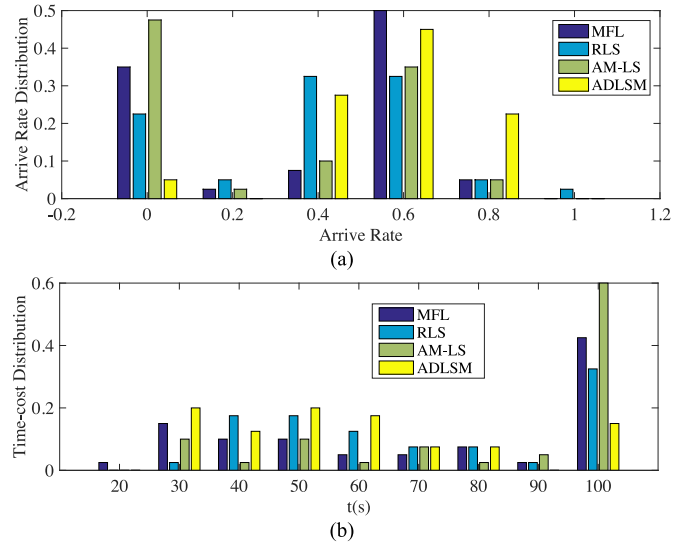


Fig. 17. Performances comparison by statistical histogram. (a) Arrive rate statistical histogram. (b) Time-cost statistical histogram.

TABLE VIII
PERFORMANCE COMPARISON

	AR	TC	AbR	ST
MFL	0.38±0.29	52.50±22.08	40%	-
RLS20	0.40±0.26	58.00±18.88	27.5%	2.98±1.53
RLS10	0.52±0.34	62.19±18.87	30%	6.85±2.61
RLS5	0.44±0.40	66.22±15.77	65%	17.48±3.90
AM-LS	0.30±0.30	59.45±24.58	55%	37.58±20.70
ADLSM	0.56±0.19	51.20±17.18	12.5%	12.10±8.98

these strategies. To further test the performance of our model, we also compare the absence rate (AbR) and ST besides AR and TC. The AbR is the percentage of experiment times in which no robot arrives at the target position.

In Table VIII, the number after the RLS is the switch period time, for instance, RLS20 means the ST period is 20 s. Table VIII shows that the ADLSM has the highest AR and the lowest AbR. Moreover, as the intuitive conclusion mentioned before, the MRS with the ADLSM has the smallest square errors of AR and TC. It can be also observed that for the RLS strategy, switching leader frequently results in a higher TC and AbR. Thus only a suitable switching strategy can improve the performance of the MRS, not simply arbitrary methods. Comparing with the AM-LS, the ADLSM needs less switching time but performs much better than it. Because the white noise assumption in the MAS does not match the real case in the MRS.

Based on the above analysis, our model has a higher probability to lead more robots to reach the target in time. This makes the formation error lower, thus enhances the adaptability of the system to the changing environment.

It can be seen from the graphs that the MRS with the ADLSM is more effective to deal with the complicated terrain. We may draw the conclusion from the previous simulation results that the proposed algorithm can yield more reliable results when facing environment disturbance.

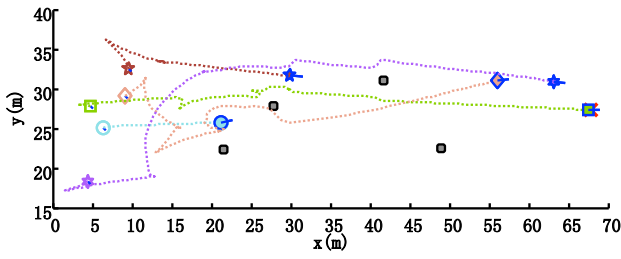


Fig. 18. Trace evolution for the MRS with leader failures.

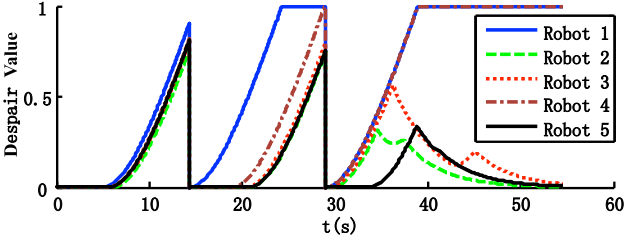


Fig. 19. Despair evolution curves for the ADLSM.

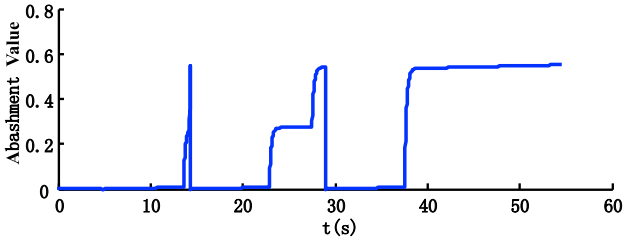


Fig. 20. Abashment evolution curves for the ADLSM.

D. Fulfilling the Mission After Leader Failure

In this case study, we consider the leader failure challenge. When a team moves toward its destination, the global leader fails to move suddenly. Since other robots' failures will not affect the whole team's cognition of their destinations, the discussion about that situation is unnecessary. It is obvious that with a fixed model, all members cannot move when the global leader fails since it is the only one knowing the destination and followed directly or indirectly by all the other robots. Thus only the MRS with the ADLSM is considered. The simulation environment is limited into a 70×25 rectangle space with obstacles. In an MRS with five robots, the current global leader (marked as a blue circle) stops at the 5th second after their trip starts. The second leader, which is marked as a purple star in this case, is down at the 20th second. The moving traces are presented in Fig. 18 and affection states are shown in Figs. 19 and 20, and the corresponding leader switching process is provided in Fig. 21.

It can be observed that the team can reselect the global leader autonomously when the leader failure happens. Consequently they can keep moving toward the target.

A notable thing is that the leader switching process and leader failure do not happen at the same moment. This is because the despair of team members cumulates over time when the global leader is in failure. As we discussed before, this delay is necessary to prevent too much switching.

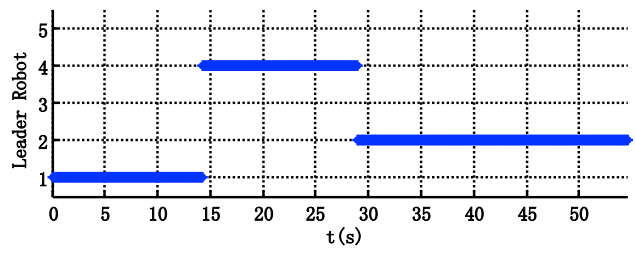


Fig. 21. Illustration of leader switching with the ADLSM.

V. CONCLUSION

In this paper, we propose a self-adaptive dynamic leader selection model based on virtual affections and successfully apply it to solve the problem of a trapped or failed leader in the formation control of the MRS. In order to describe the status of robots, an FIS is employed to construct an environmental perception module. A greedy algorithm with a swap operation is proposed to ensure that the robot team can be formed as quickly as possible after switching a leader. The proposed model switches a leader autonomously when the team falls in trouble, thereby improving the probability for the team to get out of the woods. It does not require additional information to notify team members to initiate a leader selection when the leader has a failure. Robots will select a new leader according to their affection state and continue to fulfill the mission. Simulation results demonstrate the effectiveness and adaptability of the proposed model.

In this paper, our discussion is under the assumption that the communication among robots has no limitation. How to implement cooperative control under the limited communication and switching topology is the focus of our future research. In addition, the DLS for multitarget tracking is also our interest.

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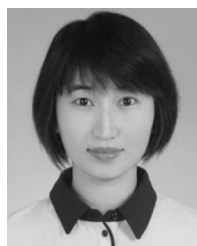
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